

INDIVIDUAL PROJECT (LSDS)

WEEK 7 REPORT

(Data Visualization, Business Impact, Limitations, and Future Scope)

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1) Introduction to Week 7:

In Week 7 of the project, data visualization techniques were applied to analyze the Car Price Dataset. The objective was to transform raw data into meaningful insights using graphical representations.

Various charts such as bar charts, pie charts, and line graphs were created using Python libraries including Pandas and Matplotlib. These visualizations help identify patterns related to car brands, fuel types, transmission types, vehicle condition, price ranges, and price trends over time.

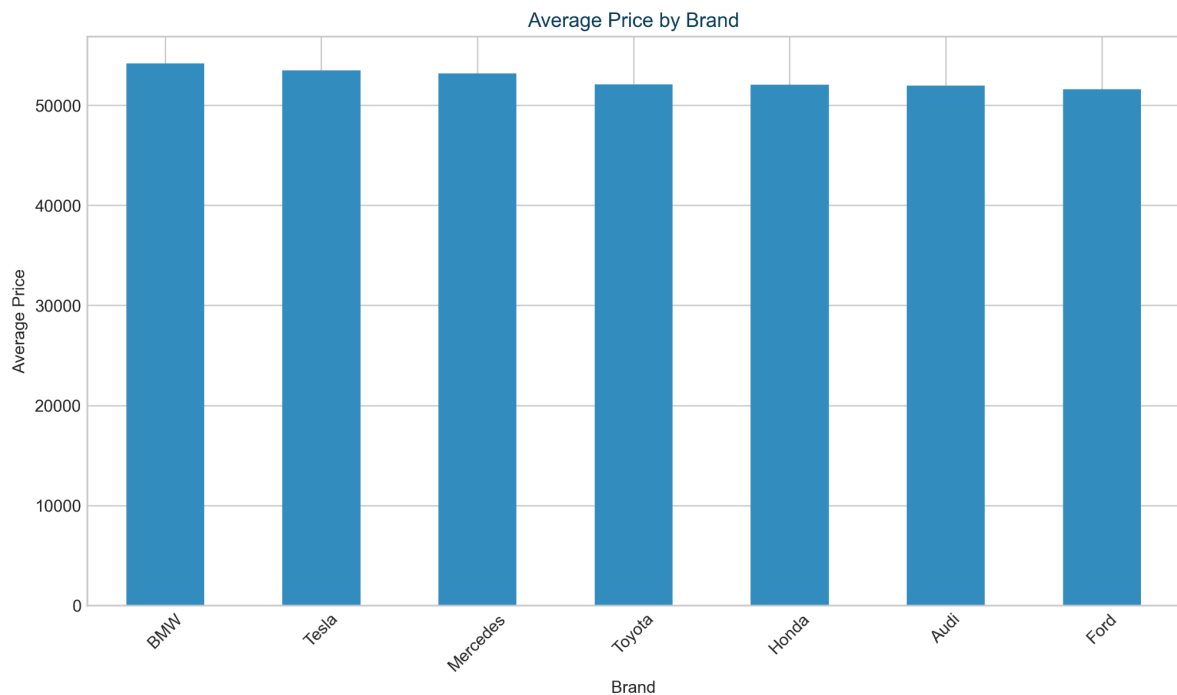
In addition to the charts, a dashboard was created to display multiple visualizations in a single view, making it easier to understand the dataset. A Hadoop ecosystem architecture diagram was also designed to illustrate the workflow of the project using different Hadoop components.

2) Visualizations Created:

2.1) Average Price by Brand:

- This bar chart shows the average price of vehicles for each car brand in the dataset. The dataset was grouped by brand and the mean price was calculated.
- Observation
- The visualization shows that BMW has the highest average price, followed closely by Tesla and Mercedes. Brands such as Toyota, Honda, Audi, and Ford

have slightly lower average prices. This indicates that luxury brands generally have higher vehicle prices compared to mainstream brands.

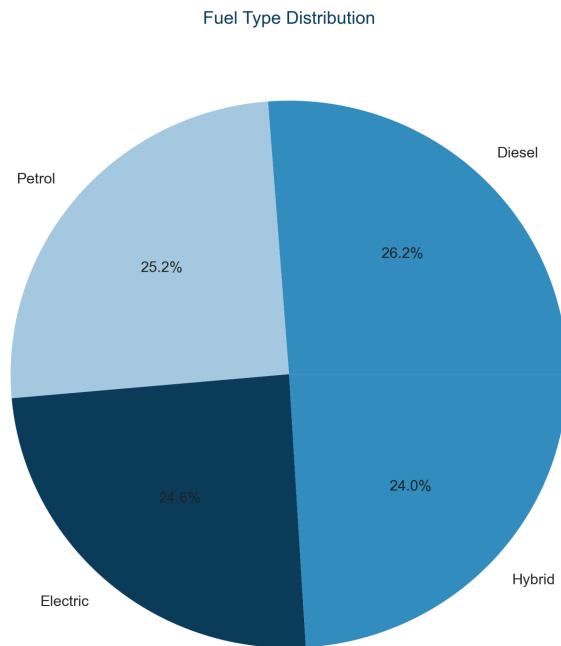


2.2) Fuel Type Distribution:

This pie chart represents the distribution of vehicles based on fuel type in the dataset.

Observation

The dataset contains four fuel types: Diesel, Petrol, Electric, and Hybrid. Diesel vehicles have the highest proportion, followed by petrol vehicles, while electric and hybrid vehicles also represent a significant share of the dataset. This reflects a mix of traditional and modern fuel technologies in the automobile market.

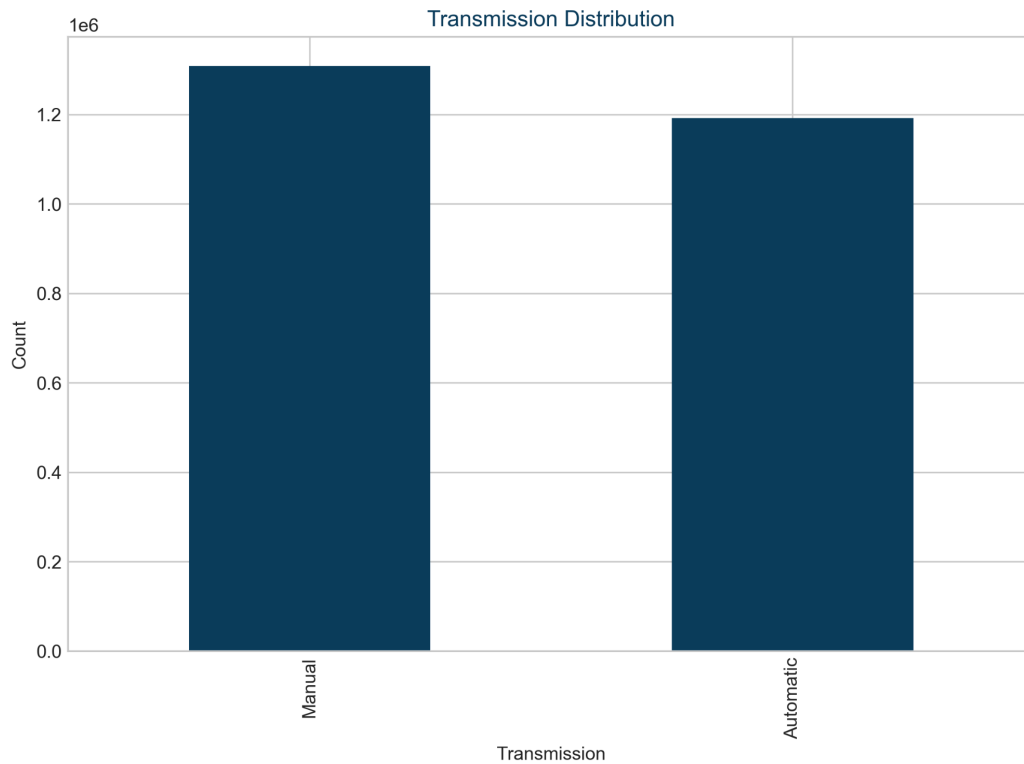


2.3)Transmission Distribution:

This bar chart shows the distribution of vehicles based on transmission type.

Observation

The chart shows that manual transmission vehicles are slightly more common than automatic vehicles in the dataset. However, automatic transmissions remain popular due to their convenience and ease of driving.

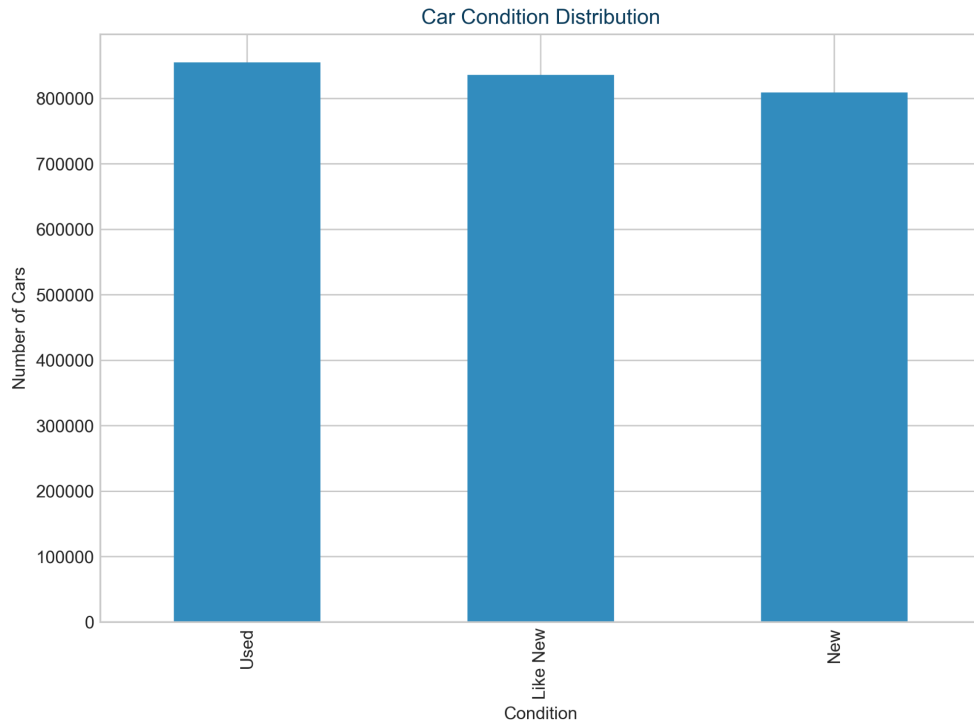


2.4) Car Condition Distribution:

This bar chart represents the distribution of vehicles based on their condition.

Observation

Vehicles in the dataset are categorized as Used, Like New, and New. Used cars appear in the highest numbers, followed by like-new vehicles and new cars. This indicates that a large portion of the dataset represents the used car market.

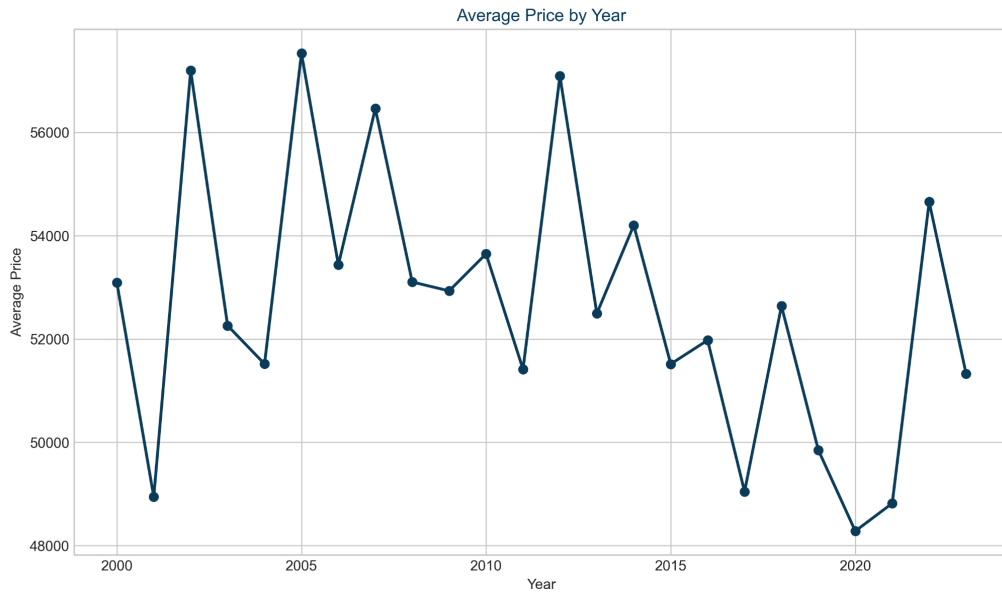


2.5) Average Price by Year:

This line graph illustrates the trend of average car prices across different years.

Observation

The graph shows fluctuations in the average price of vehicles across different years. Some years show higher average prices, which may be influenced by newer car models, improved technology, and market demand.

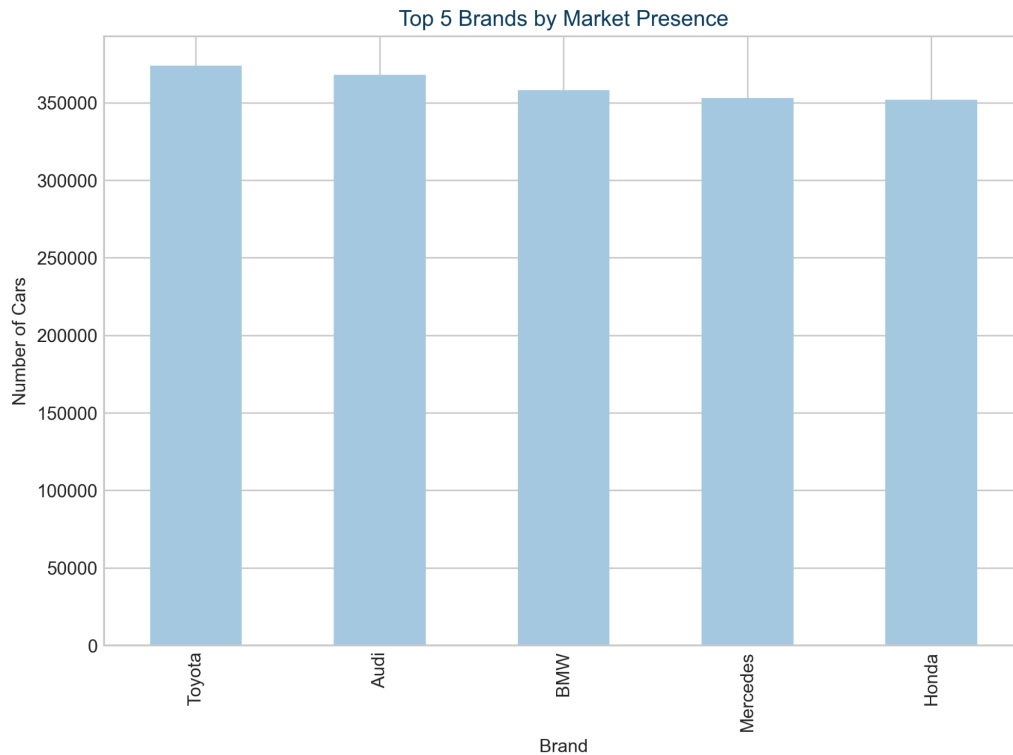


2.6) Top 5 Brands by Market Presence:

This bar chart shows the top five car brands with the highest number of vehicles in the dataset.

Observation

The visualization indicates that Toyota has the highest market presence, followed by Audi, BMW, Mercedes, and Honda. These brands appear most frequently in the dataset, suggesting that they represent a large share of the vehicles analyzed.



2.7 Price Range Distribution:

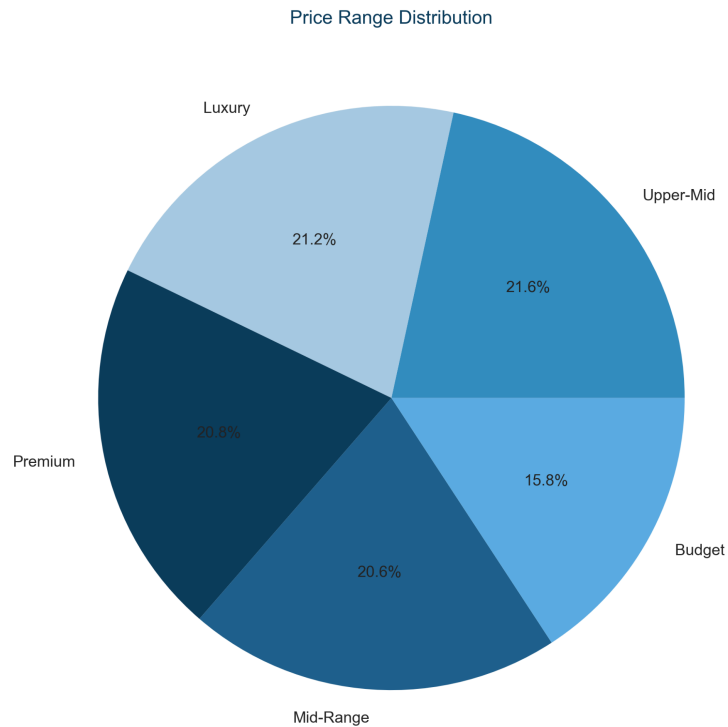
This pie chart categorizes vehicles into different price segments.

The price segments include:

- Budget
- Mid-Range
- Premium
- Upper-Mid
- Luxury

Observation

The distribution shows that a large portion of vehicles fall into the mid-range and upper-mid price categories, while luxury vehicles represent a smaller portion of the dataset.



3) Dashboard Visualization:

A dashboard was created using Python to combine multiple charts into a single visual interface. The dashboard provides an overview of the dataset by displaying several key visualizations together.

The dashboard includes:

- Average price by brand
- Fuel type distribution
- Price distribution histogram
- Average price by transmission
- Top expensive models
- Brand distribution

Benefits of the Dashboard:

The dashboard helps present insights clearly and efficiently. Instead of analyzing multiple charts separately, users can observe key patterns and trends in one place. This makes the data analysis process faster and more intuitive.



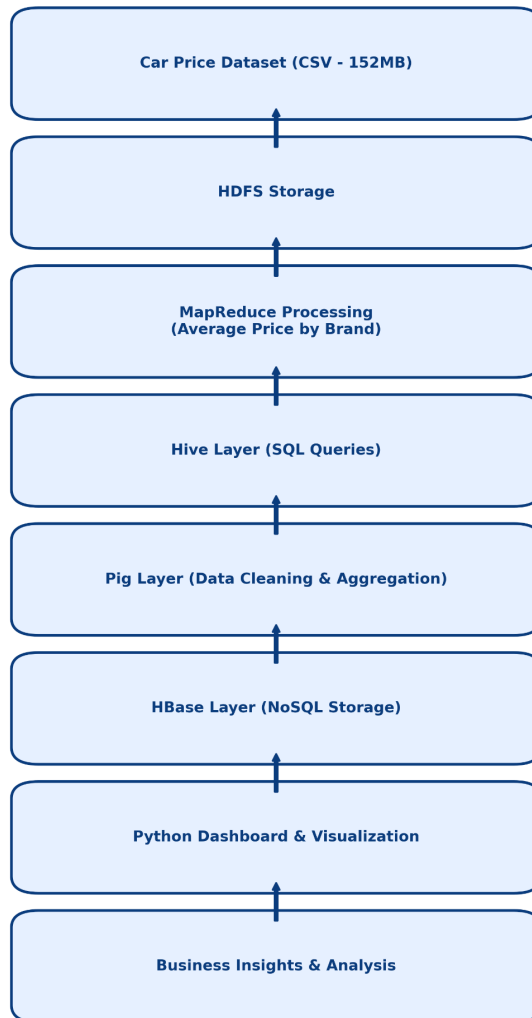
4) Hadoop Ecosystem Project Architecture:

The Hadoop ecosystem architecture diagram illustrates how different components of the Hadoop framework are used in this project.

Workflow Explanation:

1. **Car Price Dataset (CSV – 152MB)**
The dataset is initially stored in CSV format.
2. **HDFS Storage**
The dataset is uploaded into the Hadoop Distributed File System for distributed storage.
3. **MapReduce Processing**
MapReduce is used to process the dataset and perform calculations such as average price by brand.
4. **Hive Layer (SQL Queries)**
Hive enables SQL-like queries for analyzing structured data stored in HDFS.
5. **Pig Layer (Data Cleaning & Aggregation)**
Pig is used for data cleaning and aggregation tasks.
6. **HBase Layer (NoSQL Storage)**
HBase is used to store structured data in a NoSQL format.
7. **Python Dashboard & Visualization**
Python libraries are used to create visualizations and dashboards.
8. **Business Insights & Analysis**
The final stage involves extracting insights from the processed data.

Hadoop Ecosystem Project Architecture



5) Key Insights:

From the visualizations and dashboard analysis, the following insights were observed:

- BMW, Tesla, and Mercedes show higher average prices compared to other brands.
- Diesel and petrol vehicles dominate the dataset.
- Manual transmission vehicles appear slightly more frequently than automatic vehicles.
- Most vehicles in the dataset are used cars.
- Toyota has the highest representation among the brands in the dataset.

- A large number of vehicles fall within the mid-range price category.
- Vehicle prices fluctuate across different years rather than following a fixed upward trend.

6) Conclusion:

The visualizations and dashboard developed in this stage successfully transformed raw car price data into meaningful insights. By analyzing brand pricing, fuel types, transmission types, vehicle condition, and price trends, the dataset provides a clearer understanding of patterns within the automobile market.

The integration of Python-based visualization with the Hadoop ecosystem architecture demonstrates how big data tools can be combined with analytical techniques to support data-driven decision making.